International Journal of Information Security (2025) 24:197
https://doi.org/10.1007/s10207-025-01112-1

REGULAR CONTRIBUTION

Evaluation of the maturity of LLMs in the cybersecurity domain

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/ Accepted: 10 August 2025 / Published online: 27 August 2025
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Abstract
The increasing sophistication of cybersecurity threats demands innovative solutions for network defense and security education. This study evaluates the potential of Large Language Models (LLMs) in automating cybersecurity tasks, specifically focusing on three critical areas: Honeypot generation, Malware Detection, and Capture The Flag (CTF) exercise creation. We introduce two novel evaluation frameworks: the Cybersecurity Language Understanding (CSLU) benchmark, which assesses model knowledge through domain-specific multiple-choice questions, and an automated evaluation system that measures models' ability to generate functional security artifacts. Using these frameworks, we evaluated seven state-of-the-art LLMs, including GPT-4, Gemini Pro, and Claude 3 Opus. Results demonstrate that current LLMs exhibit strong capabilities in Malware analysis, with four models achieving perfect scores. However, performance varied significantly in CTF exercise generation, indicating areas for improvement. GPT-4, Gemini Pro, and Claude 3 Opus consistently outperformed other models across all tasks. Performance patterns suggest that model size correlates with task effectiveness, though architecture-specific optimizations also play a significant role. Our findings indicate that LLMs can effectively automate certain cybersecurity tasks, particularly in Malware Detection and analysis. However, their capabilities vary across different security domains, suggesting the need for specialized training or domain-specific adaptations for optimal performance.

Keywords Cybersecurity · Large language models · Honeypots · CTF challenges · Malware detection

1 Introduction

In today's global cybersecurity landscape, organisations face significant challenges in protecting their networks and systems from progressively more sophisticated threats. This accelerated technological advance and the growing dependence on interconnected systems have made cybersecurity a central component of defence strategy. From this perspective, the complex and dynamic nature of the threat ecosystem requires innovative and adaptive approaches to ensure the integrity, confidentiality and availability of sensitive information. In addition to these considerations, there is also the need to keep professionals up to date with the latest cybersecurity techniques and tools in order to implement the required approaches, which requires a constant training effort.

Large Language Models (LLMs) have gained a lot of interest in recent years due to their ability to generate coherent text and perform natural language processing tasks [9]. LLMs are artificial intelligence models that use deep learning techniques to process and understand natural language. These models are trained on large amounts of text data to learn patterns and relationships between entities in the language and thus acquire knowledge about the real world and various domains.

Given the mentioned appreciation and use of LLMs, these models have become a revolutionary element not only in our daily lives, but also in the technological and academic sectors. These innovations have transformed the way we interact with information, technology and the world around us, moulding various aspects of our lives.

Recognising the urgent need to strengthen organisations' cybersecurity capabilities [4], it is considered crucial to explore the integration of LLMs with cybersecurity systems. By leveraging advanced artificial intelligence, the aim is to

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proactively address cybersecurity challenges, providing an efficient response to constantly evolving threats.

These challenges require an innovative solution that allows organisations to remain at the forefront of cybersecurity, responding effectively to evolving threats while optimising the use of their human resources.

The objective is to evaluate the use of LLMs to automate the development of cybersecurity mechanisms, by assessing the current state of maturity in relation to the topics to be studied. The purpose of this research is to add value to the field of using LLMs in cybersecurity. The intersection between natural language processing and information security will be explored in greater depth, through the development of 2 studies aimed at evaluating the use of this type of model to automate the development of cybersecurity mechanisms.

This research specifically examines to what extent current state-of-the-art LLMs possess the domain-specific knowledge required for cybersecurity applications, particularly in Honeypot generation, Malware detection, and CTF exercise creation. The study further investigates how the performance of these models varies between different cybersecurity tasks and identifies the factors, such as model architecture and parameter size, that influence this performance. Furthermore, we address whether LLMs can autonomously generate effective and functional cybersecurity artifacts without domain-specific training or fine-tuning.

1.1 Paper structure

This article comprises six main sections, followed by an Appendix, each serving a specific role in developing and presenting the research.

Section 1 serves as an introductory chapter, it begins by introducing the research topic and providing a general overview.

The following section (Sec. 2) reviews the most recent and impactful contributions within the field explored in this article. The focus here is on the current state of LLMs, then addresses Prompt Engineering techniques, crucial for enhancing LLM performance and accuracy in generating desired outputs.

Next, Sec. 3, covers the studies conducted throughout the research. The experimental tests and procedures are presented here, with a detailed description of the methodologies employed in each experiment.

Followed by the results in Sec. 4, which the results of the experiments are displayed in a clear and organized manner, allowing for an accurate interpretation of the data collected.

Section 5 presents the analysis of the outcomes and their implications in the field.

Section 6 provides a reflective summary of the research work, synthesizing the main findings and their contribution to the field.

2 State of the art and related work

This chapter aims to situate the reader in the context of the research, establishing a solid basis for the research in question and highlighting the main trends and findings.

In order to carry out this experimental test, topics such as Prompt Engineering, benchmarks to obtain knowledge about the state of the art with the aim of structuring the study and consequently achieving the best and most robust results. Work related to investigating the use of LLMs for cyber defence mechanisms and in CTF exercises was also investigated.

2.1 Prompt engineering

Prompt Engineering is an area dedicated to the development and optimisation of prompts for generative artificial intelligence systems. It is considered an important area of computer science because it has the potential to significantly improve the effectiveness and performance of generative artificial intelligence systems. Effective prompts can help ensure that artificial intelligence systems generate accurate, relevant and useful results. There are a number of documentation projects that address the area of Prompt Engineering. One of the best known is OpenAI's Prompt Cookbook[1]. This cookbook is an open-source collection of examples and guides on how to create effective prompts for language models. This project has received a lot of contributions from various studies and published articles, such as in [16], where the authors explored an innovative technique called "Chain-of-Thought Prompting" to optimise the ability of LLMs to perform complex reasoning. This technique involves presenting examples of intermediate chains of thought to the LLMs, encouraging them to develop a structured and logical thought process. Tests carried out with three different LLMs have shown that "Chain-of-Thought Prompting" provides significant improvements in performance on arithmetic, common and symbolic reasoning tasks.

Another important documentation project is Google AI's Prompt Engineering Handbook [2]. This handbook provides a comprehensive overview of the field of Prompt Engineering, including basic concepts, advanced techniques and use cases. It's also worth mentioning the Prompt Engineering guide developed by DAIR.AI [3], which aims to democratise AI research, education and technologies.

[1] https://cookbook.openai.com

[2] https://developers.google.com/machine-learning/resources/prompt-eng

[3] https://www.promptingguide.ai

2.2 Benchmarks

In this constantly evolving spectrum of LLMs, benchmarks are indispensable tools for evaluating and comparing the performance of these models in a wide range of tasks. These benchmarks serve as standardised benchmarks, allowing researchers and programmers to map progress in the field and identify the nuances that distinguish the different LLMs.

There are at the moment some benchmarks capable of assessing the state of knowledge of LLMs in a structured and standardised way, such as Massive Multitask Language Understanding, developed by [8], which is a widely recognised performance benchmark applied to LLMs. The LLM assessment model in question is generally made up of prompts that follow a multiple-choice question/answer format, where a wide range of subjects and question types are covered, including sciences, humanities, and technical questions. In short, the benchmark described seeks to test the depth and breadth of the models' knowledge, as well as their inference capacity, by applying questions that vary in complexity and require different levels of contextual and factual understanding.

In the field of cybersecurity, although the offer of benchmarks is considerably reduced, there are some benchmarks that focus on evaluating the knowledge of models in the area of cybersecurity, an example of this type of benchmarks is CyberBench, where [11] used five tasks to evaluate a set of domains, and was designed to evaluate the performance of LLMs in tasks related to cybersecurity. The tasks adopted by the authors are: named entity recognition, summarisation, multiple choice and text classification. The summarisation task made up of two datasets, the Computer Security MMLU (SecMMLU), which is a subset of the MMLU focused, in general terms, on the cybersecurity domain, and the CyQuiz, which follows the same question format as the previous one. This article demonstrates the results of using CyberBench to evaluate more than ten LLMs. It is worth highlighting the interesting results obtained by the models trained specifically for the evaluation of this article, which were able to obtain the best classification in some tasks.

Although published this year, most of the companies that own the models studied have already presented improved versions of them and there was also a lack of models capable of competing with the favourite models in terms of evaluation following the MMLU parameters in this experimental test, such as Google's Gemini or Anthropic's Claude 3.

The authors of SECURE, [2], have introduced a benchmark designed to assess the performance of LLMs in realistic cybersecurity scenarios. SECURE includes six datasets focussed on the Industrial Control Systems sector, which seeks to assess knowledge extraction, comprehension and reasoning based on industry-standard sources. The study evaluated seven models, and generated insights such as recommendations for improving the reliability of LLMs as cybersecurity consulting tools.

2.3 Cyber defence mechanisms

Given the current technological modernisation, cyber defence mechanisms are considered essential to protect networks and systems against threats in cyberspace. These include technologies and strategies designed to prevent, detect and respond to attacks [4]. A concrete example of cyber defence strategies are deceptive techniques. In a corporate environment, deceptive mechanisms refer to security techniques that aim to deceive or confuse potential malicious actors in order to protect a company's assets. These techniques include the creation of traps, false information, or the simulation of vulnerabilities in order to discourage or delay intrusions. Examples of deceptive mechanisms include Honeypots or honeynets.

A Honeypot is a computer security technique that aims to attract potential cyber-attacks by simulating a legitimate, vulnerable resource in order to monitor and analyse attackers' tactics, techniques and procedures. This mechanism consists of exposing an intentionally unprotected system or service configured to attract malicious activity, allowing security professionals to assess emerging threats and strengthen defences in real time.

[15] developed an effective isolation prediction and maintenance system for cloud security and Intrusion Detection Systems (IDS). The study compared different levels of Honeypots to understand their details and interaction with the network. A real-time Honeypot was used to detect attacks on a SSH server.

Given the ability of LLMs to recognise patterns, it would be interesting to consider using them to develop new approaches to Malware Detection. For example, an LLM could be trained on a dataset of malicious and legitimate software. This would then make it possible to use a model to identify patterns in the code that are characteristic of malicious programmes. This approach is intended to have several advantages over traditional Malware Detection methods. Initially, they would be more effective in detecting unknown Malware, as LLMs can learn to identify patterns, and even recognise obfuscation patterns, that are common to different Malware families, even if those families are not known, and even if they do not initially identify a malicious pattern, they can be dynamically updated and adapted as new threats emerge. This allows for a faster response to changes in the threat landscape, while maintaining detection effectiveness. This type of model is less susceptible to false positives, as it learns to identify patterns that are specific to Malware, resulting in a reduction in the likelihood of a legitimate programme being identified as malicious. Another advantage would be

that, given the multi-modality of these models, they could also be used to identify Malware in different formats. This would make the analysis process more effective, for example in cases of steganography where relevant information or even malicious code is hidden in files of other formats, such as images.

[14], having identified that many of the current Honeypots solutions lack the realism needed to engage and deceive human attackers, presented an innovative Honeypot system, based on LLMs, which showed remarkable results in achieving an accuracy rate of 92% in assessing its ability to engage and deceive human attackers with dynamic and realistic responses. [12] also studied the use of ChatGPT as an innovative tool for creating Honeypots interfaces in cybersecurity. The article presents ten different tasks that the language model can appropriately respond to an attacker using line commands. The usefulness of this Honeypot depends on its ability to delay or prevent attacks on critical network assets, such as databases or confidential information. The authors concluded that ChatGPT's ability to identify and mitigate malicious activity makes it a viable tool for organisations looking to optimise their cybersecurity posture.

In the Malware analysis domain, [7] explored active learning techniques for managing a non-stationary model for Android Malware Detection over a seven-year period. It assessed the impact of various elements, including feature space selection, data balancing, and timestamping methods, on the model's performance. The authors compared detection accuracy and labeling costs with baseline models, also examining model resilience to noisy labels caused by human error or adversarial interference. By focusing on active learning in non-stationary Malware Detection, this study addresses a key gap in existing literature.

From a different perspective of cyber defence, [5] presents a study on the use of an LLM for Malware Detection in an obfuscated code context. As did [19] who, using the ChatGPT-4 model, generated security tests in order to detect malicious third-party libraries, also demonstrating how vulnerable library dependencies expose certain applications to supply chain attacks. Like [6], the authors recognise the growing expectation of its effectiveness in detecting vulnerabilities from LLMs. However, a quantitative understanding of its potential in this area was still lacking. The authors therefore introduced VulBench, a vulnerability benchmark. This benchmark gathers high-quality data from a variety of CTF challenges and real-world applications. Through tests involving 16 LLMs and 6 state-of-the-art deep learning models and static analysers, it was found that several LLMs outperform traditional deep learning approaches in detecting vulnerabilities, revealing untapped potential in LLMs. This work contributes to the understanding and use of LLMs to optimise software security.

2.4 CTF exercises

The CTF exercises, short for Capture The Flag, are cybersecurity challenges where participants can compete against each other. These exercises are designed to test and enhance in a dynamic and challenging way not only cybersecurity-related skills, but also creativity and problem-solving skills. These challenges can range from simple to complex, and can cover a wide range of topics, such as analysing Malware, reverse engineering, steganography, cryptography, communication networks, exploiting vulnerabilities, digital forensics, and much more.

Nowadays, CTFs are an important part of cybersecurity culture and are widely used in academic, business and community contexts. This type of exercise has recently gained prominence in the research field, for example [10] used eleven CTF exercises on Heap vulnerabilities to evaluate their proposed solution. [1] developed a deceptive mechanism to understand the behaviour of human attackers in order to optimise the defender's strategy, used CTF environments on vulnerable applications in order to collect data on attacker behaviour, rather than artificially generating data based on accepted assumptions about opponents.

The research into the integration of LLMs regarding the creation of CTF exercises represents a promising area. By understanding the ability of these models to simulate real situations and generate diverse challenges, they can significantly increase the quality and quantity of competitions.

By integrating LLMs into the development of exercises, the creators themselves will be able to save time, guarantee technical quality, and offer more complex and engaging challenges.

3 Experimental methodology and evaluation framework

This section presents the comprehensive experimental methodology employed to evaluate LLM capabilities in cybersecurity domains. Our approach consists of two complementary evaluation frameworks: (i) a benchmark assessment of theoretical knowledge through the Cybersecurity Language Understanding (CSLU) benchmark; and, (ii) our own autonomous evaluation system for assessing practical artifact generation capabilities. We first detail the models selected for evaluation, followed by the design and implementation of each component.

3.1 Models

The process of choosing the models to include in the studies was based considering the models used, but not exclusively, in other studies analysed in section 2, through factors such

Table 1 Models to evaluate

Model's name	Parameterisation	Company
GPT-4	1.76 trillion	OpenAI
GPT-3.5	175 billion	OpenAI
LLama 3	70 billion	Meta
Code LLama	70 billion	Meta
Claude 3 Opus	2 trillion	Anthropic
Gemini Pro	N.A	Google
Mixtral	8 x 22 billion	Mistral

as the visibility of the models, the date of launch of them or their respective parameterisation, since this parameterisation shapes the understanding of the model's language. It is therefore assumed that the more a model is parameterised, the more likely it is to achieve better results. Ultimately, 7 models were chosen; the best models from widely recognized LLM companies were chosen to maximize the scope of the evaluation.

Thus, the models used in this experimental study are available in the table 1, for each model there is also attached information about its parameterisation and the company that currently owns the model. Unfortunately, at the time of submitting this document, it was not possible to acquire any information on the parameterisation of the Gemini Pro model.

Recognising the accelerated development of this type of model, it is assumed that on the date of submission of this thesis, some of the models used do not represent the model with the best capabilities presented by the company that owns them.

3.2 Cybersecurity language understanding

In order to assess the viability of the models in terms of the theoretical and practical knowledge of the objects of study introduced, a benchmark was developed. This first experimental study aims to assess the current state of knowledge of the models to be evaluated, in order to establish a starting point for this research.

This benchmark was based on the MMLU benchmark [8], following the model evaluation process used by the authors, using multiple choice questions. In this format, the models are presented with a specific problem and given four options to choose from, only one of which is correct. To ensure a complete evaluation of the models, 10 questions were created and submitted for each topic.

The benchmark developed, called Cybersecurity Language Understanding (CSLU), offers a standardised evaluation framework, guaranteeing consistency and reliability in the results obtained. By implementing this benchmarking method, it is possible to assess the quality of the models evaluated from a capabilities perspective on cybersecurity topics.

Table 2 Question example - Honeypot

Which of the following CVEs is the version 1.23.1 of Nginx vulnerable of?
A) CVE-2022-41741
B) CVE-2021-23017
C) CVE-2019-9513
D) CVE-2019-9511

Given the fact that the evaluation of the models in this experimental test is carried out through multiple choice questions, taking advantage of the analytical nature of the results data, the main evaluation metrics were considered to be the effectiveness rate, the average value and the standard deviation.

3.2.1 Prompts

The format of prompts using multiple choice questions facilitates binary classification in the evaluation process and provides a clear and measurable way of assessing the performance of the different LLMs, ensuring that their capabilities can be directly compared and analysed. The prompts were mostly run through the interfaces provided by the companies that own the models, with the exception of the Meta and Mistral models, which were run through Perplexity Labs' accessible interface.

The Tables 2, 3 and 4 show concrete examples of the prompts used to evaluate the models for each topic. In each figure, the correct answer is marked in bold.

As shown in figure 2, the prompts for the topic of Honeypots are intended to assess knowledge of the models in question, both in theoretical terms and in more practical approaches. This content spectrum aims to assess everything from the theoretical terms of the Honeypots topic to knowledge of the documented presence of vulnerabilities in globally known commercial products. The figure in this topic aims to verify the state of knowledge of the models regarding the catalogue of vulnerabilities present in Nginx servers.

The focus of the CTF exercises is to determine the state of problem-solving skills and general knowledge of cybersecurity models. Theoretical mastery of the content to be generated and problem-solving skills are the central points involved in CTF exercises. Given this fact, ensuring that the models have both characteristics is essential in order to ensure their viability in generating this type of artefact. The prompt in the figure 3 challenges the models by solving a simple CTF exercise. This question is considered to be part of the feasibility study on the generation of CTF exercises, since it

Table 3 Question example - CTF

What is a more reliable approach to determine the Caesar Cipher key used for encryption, rather than simply guessing a common shift value?
A) Try all possible key values (0-25) and see which decrypted message makes the most sense
B) Look for patterns in the letter frequencies of the ciphertext
C) Both a) and b) are useful techniques
D) There's no reliable way to determine the key without additional information.

Table 4 Question example - Malware

Arrange the following steps in the correct sequence of a keychain attack:
1) The backdoor grants the attacker remote access to the victim's system.
2) Attacker sends a phishing email with a malicious attachment.
3) Attacker gains access to the victim's keychain or credential storage.
4) Victim unknowingly executes the malicious attachment, which installs a backdoor on the system.
5) Attacker retrieves login credentials stored in the keychain.
A) 1 - 2 - 3 - 4 - 5
B) 3 - 1 - 2 - 4 - 5
C) 2 - 4 - 1 - 3 - 5
D) 1 - 3 - 4 - 2 - 5

Table 5 Prompt engineering technique application example

Version	Prompt
Base	Analyse the following code:
CoT first phase	Analyse the following Android application:
CoT second phase	Analyse the following Android application, before performing the analysis consider the standards and definitions of every Malware category within the Android ecosystem:

Table 6 Evaluation prompt example

You are a LLM evaluator, I will provide you some LLMs prompts and the respective responses from the models, you will have to classify the models responses into 3 groups: correct, plausible and incorrect. The correct classification only applies to the responses that fully accomplish the requested features or needs of the prompts, if the response contains all needed information and resources to the developer implement the challenge. The plausible classification applies to the responses that didn't fully accomplish the requested of the prompts, for example if the model didn't provided all necessary tasks to complete the CTF if needed. The incorrect classification only applies to the responses that fully didn't accomplish the requested features or needs of the prompts.
User prompt: "%s"
Model response: "%s"

is first necessary to ensure the theoretical and practical consistency of the knowledge spectrum involved in this type of exercise.

The study on Malware aims to assess the ability of models not only to identify and analyse the presence of threats in software, but also theoretical terminologies on this subject. Although in a real context it is expected that the analysis of suspicious artefacts will focus on binary files, the aim of this study is to assess the state of theoretical and practical knowledge in which it is understood that the use of Malware source code satisfies this assessment requirement. The prompt in the figure 4 asks the models to organise certain events in the correct sequence of a Keychain attack.

3.3 Autonomous evaluation of LLMs

A second experimental study was carried out with the aim of identifying the best models in terms of their viability in cybersecurity applications, introducing them to tests that are contextually closer to real scenarios and use cases. The responses of the LLMs will be evaluated automatically using an additional LLM. Consequently, the aim of this evaluation

is not to develop an automatic workflow or benchmark, but to identify the best models without the need of user's analysis or judgment of the models responses.

The same topics studied and present in the assessment carried out in the 3.2 section were considered: generating CTF exercises, generating cyber defence mechanisms such as *Honeypots*, and finally detecting *Malware*.

For this experimental test, we used the interfaces provided by the companies themselves, such as ChatGPT for GPT-3.5 or Google Gemini. The other models were accessed via the Perplexity Labs Playground.

Quantitative and qualitative facts about the artefacts generated by the models were considered as metrics for evaluating them, so as to enable a consolidated and standardised evaluation. The evaluation points adopted were whether the model successfully achieved the outlined objective, the quality of the final response and finally the number of prompts required to achieve the desired result. Taking the evaluation parameters into account, the model's answers will be classified into one of the following categories: Correct, Plausible and Incorrect.

The 'Correct' classification characterises answers that fully comply with the objectives imposed by the prompt, i.e. if the answer presents all the information and resources necessary for the programmer to implement or generate the desired artefact. The 'Plausible' classification applies to answers that do not fully comply with the request, but coherently present all the steps needed to achieve the desired objective, for example, if the model does not provide all the resources needed to implement a CTF but describes the process of doing so in sufficient detail. Finally, the 'Incorrect' classification applies only to answers that did not fully fulfil the characteristics or needs requested in the prompts.

The best model will be the one that first achieves the highest accuracy values in the 'Correct' category. For tie-breaking purposes, the level of accuracy of the answers classified as 'Plausible' will be taken into account, and consequently the third and final category.

3.3.1 Prompts

For each topic, 4 to 5 prompts were conceptualised. These are organised in order of complexity, with the first representing the simplest task and the last presenting the most technically complex or ambiguous task. If a model does not achieve the expected result with a given prompt, some Prompt Engineering techniques were used to optimise the model's performance, such as Chain-Of-Thought Prompting, which allows for more complex reasoning resources through intermediate inference stages. For example, the Table 5 specifically demonstrates the use of Chain-Of-Thought Prompting in two phases, thus showing the evolution of this prompt and subsequent possible changes to the introductory paragraph it has undergone in order to optimise the LLMs' inference when necessary.

Generating CTF exercises can be a tricky task, as many CTF challenges involve advanced cybersecurity concepts, as well as the creative and authentic process required. LLMs may not have the necessary knowledge of these concepts, which can result in the creation of challenges that are not technically accurate or challenging enough for experts in this field. Determining the appropriate level of difficulty for a CTF challenge is crucial. LLMs can also find it difficult to properly assess the complexity of a challenge, resulting in exercises that are either too easy or too difficult, ideally these should be challenging, but they should also be achievable. Diversity is also key in CTF challenges to keep participants interested, creating a variety of personalised and unique challenges can be difficult for language models as they tend to follow existing examples through the standards studied. The main aim is to assess the creativity of the models; initially, objective and deterministic prompts were developed, requiring an exercise on a more specific theme or area, compared to the others, which were a little more ambiguous and lacked detail.

When building threat detection systems such as Honeypots, the careful choice of prompts can guide the creation of effective mechanisms. When asking an LLM to generate such systems, depending on the intended objectives, specific or more general prompts are both valid approaches. A general approach reflects the request to create a Honeypot by simply describing some of the characteristics or objectives sought with the Honeypot. In this way, the first two prompts follow the ideology of this general approach. On the other hand, a specific approach consists of carrying out the aforementioned instantiation or development request for a Honeypot, necessarily imposing the use of a specific server or vulnerability. The aim is to study not only the creativity of the models, but also their adaptability through the approaches they take for each case study, depending on whether they are objective or a little more ambiguous.

Despite the potential advantages of creating LLM-based Malware Detection systems, they currently represent a challenging and complex topic in the field of cybersecurity. For this reason, it is important to study the effectiveness of this group of models in order to understand their viability in this field. Although LLMs are powerful natural language processing tools, they face several difficulties when it comes to identifying malicious software. One of the main limitations of LLMs in detecting Malware is related to the highly technical and dynamic nature of cyber threats. Today's malicious programmes are often designed to avoid detection; malicious agents use advanced obfuscation techniques. From this perspective, LLMs may find it difficult to understand the specific technical nuances of these threats, since their mastery is more centred on linguistic patterns and semantic contexts. Due to

limitations imposed by the interfaces used, in which it is not possible to present binary files to analyse the presence of Malware, in this experimental test only the analysis of Malware through its source code will be considered. Although in this scenario the evaluation is abstracted from a more realistic context, evaluating Malware Detection by analysing the source code can generate valuable insight in the sense of scientific valorisation where this study is applied.

3.3.2 Responses evaluation

While in the previous experimental study it was possible to simplify the process of evaluating model responses by applying a binary classification, in the present experimental study it is not intended to influence or restrict the range of model responses. Therefore, the evaluation of model responses was conceptualised so that it could be carried out automatically, using an LLM. Simultaneously with the process of developing the regular prompts, evaluation prompts were also developed, one for each evaluation topic, for the evaluator model to effectively classify the models' responses. The model chosen to fulfil the role of evaluator was GPT-4o. As mentioned above in the subsection 3.3, it was necessary to delimit the linguistic understanding of the evaluator model through the *prompts* used, in order to atomically perform a good evaluation of the other LLMs' answers.

The Table 6 shows an example of a prompt developed for the evaluator model to classify a response from an LLM, in this case a *prompt* to evaluate the generation of a CTF exercise.

4 Results

This section aims to present and synthesise the results obtained from the experimental tests carried out on the two LLM evaluation mechanisms. In addition to summarising them, it is also intended to provide graphical representations in order to help interpret the data generated.

4.1 CSLU

The results obtained by the CSLU highlighted a clear prominence of knowledge in relation to the topic of Malware, with four models achieving the maximum score in this task, obtaining an average hit rate of 95.7% and consequently the lowest standard deviation values, as can be seen in the figure 1.

In the CTF task, the Claude 3 Opus model showed the best performance, while the other models achieved lower scores. Although the results for this topic show a greater dispersion of results, with an average hit rate of 80%, it was the Honeypots topic that obtained the highest value in terms of standard deviation, demonstrating a greater variation of results in relation to the average.

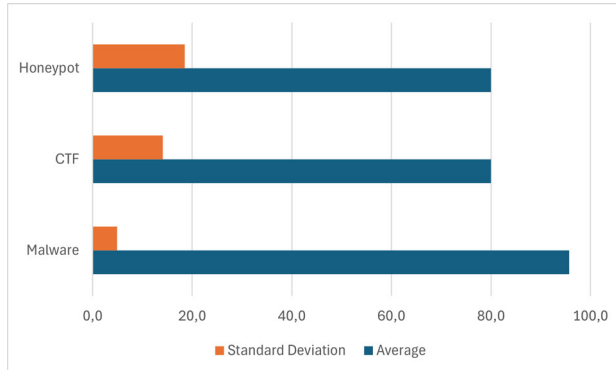

Fig. 1 CSLU results by subject

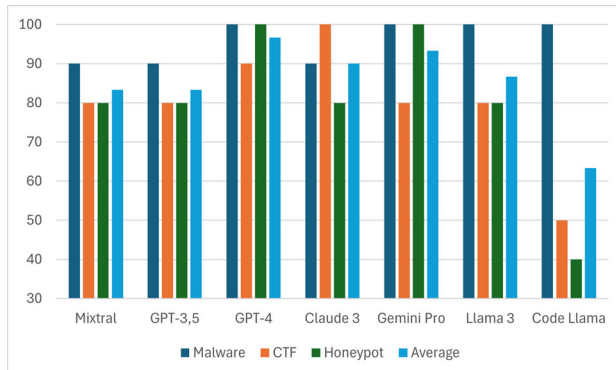

Fig. 2 CSLU results by model and subject

Overall, as shown in the table 1, it was expected that the GPT-4 and Claude 3 Opus models would be the main candidates for obtaining the best results, since these two models contain the largest number of parameters of the models studied. It was subsequently found that the GPT-4, Gemini Pro and Claude 3 Opus models showed uniformly superior results among the models studied. As can be seen in figure 2, the results obtained for each model evaluated in relation to the respective topics studied, together with the average evaluation obtained by each model.

The Code LLama model had the worst results, mainly showing difficulties with the CTF exercises and Honeypots topics. However, unlike the topics mentioned above, this model was one of the four models to answer all the questions on the Malware topic correctly, giving rise to a substantially higher standard deviation than the other models.

4.2 Autonomous evaluation

The models performed best in the task of detecting Malware, with practically all the models correctly detecting malicious programs and categorising them according to their behaviour.

Table 7 Heat map of the results obtained in the automatic test

Modelos \ Prompts	CTF					Honeypots				Malware				
	P1	P2	P3	P4	P5	P1	P2	P3	P4	P1	P2	P3	P4	P5
GPT-3,5														
GPT-4														
Llama 3														
Code Llama														
Google Gemini														
Mixtral 8x7b														
Claude 3 Opus														

The greatest inefficiency was detected in the generation of CTFs exercises; most of the models did not achieve the maximum classification due to the lack of artefacts needed to carry out the exercise, since some of the APIs used only return text. Although some models got around this limitation, for example by providing the code needed to generate these artefacts. In the case of prompt 4, which requires the generation of a CTF exercise on the subject of steganography, the models were unable to generate and return images, which consequently meant that the exercise could not be fully generated.

Overall, the GPT-3.5, GPT-4 and Claude 3 Opus models obtained the best results, notable for the fact that they only failed to obtain the maximum classification in a single prompt.

A heat map was developed to facilitate the graphic visualisation of the results obtained, consisting of three categories represented by colours. The categories used to classify the models' answers, namely "Correct", "Plausible" and "Incorrect", were thus translated into the following graphic representations: green, yellow and red, respectively. The table 7 shows this heat map.

5 Discussion

The results obtained in the subsection 4.1, from a general perspective, show that most of the models evaluated have a good level of maturity, resulting in good indicators regarding the feasibility of using these models in cybersecurity applications.

The findings of this study suggest that a possible LLM approach for cybersecurity activities could be highly beneficial, as the results are quite satisfactory in terms of the knowledge of the models evaluated in the theoretical domain of the domains studied.

However, it is recognised that there is a difference between the *benchmarks*, including the CSLU proposed by us, and the real world. Most of them lack human validation, which has led the market to find solutions, such as Arena LMSYS [3, 20], to encompass real-world cases. In our case, it is therefore necessary to explore ways of making these cases representative in the context of cybersecurity and to align the results with the performance expected in real-world scenarios.

It was found that the biggest factor inhibiting results essentially comes from the fact that only commercial models were used, which in addition to not being specifically trained for the topics studied, themselves use defence mechanisms that can negatively influence the results. It is recognised that the use of LLMs specifically and extensively trained for a single topic would leverage the results.

The results obtained in the automatic test showed good indicators for studying the viability of these models. By studying factors such as creativity, adaptability or even the state of knowledge of the models, it was possible to identify the most promising models.

From a global perspective, after studying and analysing the results obtained, it is possible to state that automating the development and instantiation of cybersecurity artefacts through the use of LLMs is expected to provide a faster and more efficient response to emerging threats. The ability to quickly generate and deploy adapted defence mechanisms, such as Honeypots with Malware analysis and detection capabilities, will enable timely detection and mitigation of potential intrusions, significantly reducing the risk of critical systems being compromised.

Secondly, automating the generation of CTF exercises aims to increase the quality and quantity of practical materials on the subject of cybersecurity training, since not only trainers but also teachers are obliged to devote a significant amount of their time to creating statements and support and assessment materials. This approach would also make it possible to atomise assessment materials for students, helping to combat plagiarism during assessment.

In the same way, the automation of repetitive and time-consuming tasks frees up human resources, allowing cybersecurity professionals to focus on strategic analyses and the development of proactive security policies. This optimisation of human resource allocation will contribute to a significant improvement in operational efficiency and incident response capacity.

5.1 Practical applications in operational cybersecurity

The experimental results presented in this study have many direct applications in cybersecurity related operational environments. This section outlines, possible, specific implementation forms and operational considerations for organizations seeking to leverage LLMs within their security infrastructure.

5.1.1 Integration with security operations centers (SOCs)

Malware detection capabilities demonstrated by GPT-4, Claude 3 Opus, and Gemini Pro models can be used within SOC environments through several practical implementations. In our findings, we suggest that these models can serve as an effective initial triage system, that, for example, process suspicious code samples before human analysis. The accuracy rates observed in our experimental evaluation suggest that such systems could potentially reduce analysts' workload, allowing security professionals to focus on more complex threats that require human expertise.

LLMs can be deployed in order to generate automated reports for identified malware samples. Such reports would include detailed analyses of code behaviour, potential impact vectors, and recommended remediation strategies, thus enhancing the efficiency of incident response processes. The contextual understanding capabilities demonstrated by the top performing models enable them to provide meaningful analysis of anomalous events when integrated with existing Security Information and Event Management (SIEM) platforms via standardized API connections.

For the implementation within an operational environments, organizations should establish first a verification method. We recommend that LLM outputs should be validated against established detection systems during an initial calibration period. This verification process ensures reliability while allowing the organization to build confidence in the LLM-based detection mechanisms before full operational deployment.

5.1.2 Honeypot deployment and management

The capability of Honeypot generation using LLMs, provides leverage for several practical applications within enterprise security architectures. Advanced LLMs are able to "understand" system vulnerabilities, and attacker methodologies enables automated configuration of dynamic honeypots that can adapt to emerging threat patterns, representing a significant advancement over traditional static honeypot deployments.

The capability of LLM-based systems to improve their output would allow honeypots to evolve based on attacker interaction data. As threat actors engage with these deception systems, the collected behavioural data can be fed back into the LLM to generate increasingly convincing decoys. This creates a self-improving security mechanism that becomes more effective over time.

Another valuable application is the generation ofbreak organization-specific honeypots that mimic internal systems with greater fidelity by incorporating organizational context into the prompt engineering process. Security teams can develop honeypots that replicate proprietary systems and applications, creating more convincing deception environments that are more likely to engage sophisticated attackers.

Implementation of such system would typically involve a modular architecture where the LLM component generates honeypot specifications (such as using Dockerfiles) that would subsequently be instantiated through configuration management platforms, such as Ansible or Terraform. This approach allows for scaling honeypot deployment across distributed networks with minimal human intervention, thus extending defensive coverage across complex enterprise environments.

5.1.3 Cybersecurity training and workforce development

The CTF exercise generation capabilities presented in our study can be systematically applied to address critical challenges in cybersecurity education and professional development. Our results show that LLMs are able to generate training programs that could possibly adapt exercise complexity to individual skill development trajectories. This capability ensures continuous education systems that generate new challenges based on emerging threat vectors, providing security professionals with a form of keeping their knowledge up to updated with the latest attack methodologies.

In addition to providing training, the automated assessment capabilities of LLM-generated CTF exercises provide mechanisms for also evaluating cybersecurity skills with customized feedback. This approach ensures more frequent assessment intervals without increasing instructor workload. One possible result of this ability would be to create large-scale security competitions and organizational training initiatives, where unique challenges for numerous participants are created on demand.

5.1.4 Implementation considerations and limitations

While our findings show significant potential for operational deployment, several important considerations should guide implementation decisions. Human verification remains essential, particularly when analysing new threats or protecting critical systems. Rather than viewing LLMs as replacements for human analysts, organizations should conceptualize them as augmentation tools that enhance human capabilities through initial analysis, pattern recognition, and context provision.

Additionally, regular validation against ground-truthbreak datasets is necessary in order to maintain performance standards over time. As threat landscapes evolve, periodic revalidation ensures that the models' detection and generation capabilities remain effective against current attack methodologies. This validation process should include both known threat samples and synthetic examples designed to

test the limits of model capabilities. Should also be taken into account future LLM upgrades, allowing for evolution, following current market trend, and improving on new model capabilities.

Finally, it is important to acknowledge that performance may vary when applying these models to highly specialized environments, not considered in our analysis. Organizations with other security requirements or specific technology stacks should perform targeted validation using domain-specific examples before full-scale deployment.

5.1.5 Deployment architecture

The proposed integration of LLMs in operational cybersecurity follows an internal workflow model where security professionals utilise these models as assistive tools within existing security infrastructures. In this architecture, LLMs operate as backend components triggered by automated detection systems or manual requests from security analysts.

A possible workflow, for honeypot deployment, would start with an Intrusion Detection System (IDS) or Security Information and Event Management (SIEM) platforms that would detect suspicious activity patterns. A trigger would then request the LLM system for honeypot specification generation adapted to the detected activity. The LLM would then generate configuration files and deployment scripts that are would be then validated by the security operations centre personnel before instantiation through existing infrastructure management tools.

Similarly, in malware analysis scenarios, suspicious code samples, identified through: i) automated scanning; or, ii) analyst investigation; are submitted to LLMs for initial classification and analysis. The generated reports would serve as a preliminary assessment that would feed subsequent human analysis rather than replacing expert evaluation.

This internal deployment proposal, ensures that LLMs operate within a controlled environment where inputs are originated from trusted sources and outputs undergo validation before operational use.

5.1.6 Quality assurance and anomaly management

Our evaluation identified specific quality concerns in LLM-generated security artefacts that require systematic attention. CTF exercise generation demonstrated particular challenges, including incomplete vulnerability implementations that appeared functional but lacked actual exploitability, inconsistencies between challenge descriptions and technical implementations, and unintended solution paths that bypassed intended learning objectives. Such anomalies could create misleading training experiences and compromise educational effectiveness.

Similarly, honeypot configurations occasionally contained logical contradictions or inappropriate security settings that might expose systems to additional risk rather than enhancing defensive capabilities. These quality issues necessitate the implementation of automated verification systems to detect common anomaly patterns before artefact deployment, including syntactic validation of generated code, logical consistency checks for security configurations, and functional testing of generated exercises.

5.2 Limitations and future directions: specialized training approaches

While our evaluation framework focused on general-purpose LLMs a specialized fine-tuning methodologies and hybrid human-in-the-loop approaches will expectedly yield superior performance for domain-specific cybersecurity applications. The choice for a general-purpose LLM was deliberate, aimed at assessing baseline capabilities accessible to organizations without a specialized AI development infrastructure.

Recent research shows that domain-specific fine-tuning can achieve substantial performance improvements. For example, SecureFalcon, a specialized models, reaches 96% detection accuracy, as presented by [18] and multitask approaches like MSIVD, by [17], is able to achieve F1 scores of 0.92. Human-in-the-loop methodologies also show comparable, or even superior, performance in complex cybersecurity scenarios as shown by [13].

We consider that future research should systematically compare domain-specific fine-tuning performance improvements against associated implementation costs and resource requirements. Human-AI collaboration patterns also represent a promising direction for bridging automated efficiency with specialized domain expertise.

6 Conclusion

A series of experimental studies were presented and discussed in order to determine the level of feasibility of LLMs in relation to the tasks listed in the cybersecurity theme, with emphasis on solutions for generating Honeypots, Malware and CTF exercises. The adoption of structured assessment models ensured consistency and reliability in the results obtained. A systematic evaluation framework was developed and implemented to assess LLM capabilities across cybersecurity domains, providing empirical evidence for the research conclusions.

The conclusions drawn made it possible to carefully choose the most suitable models to integrate with the tool to be developed, one of the main objectives of which is to automate cybersecurity processes, particularly those that were the subject of the study. The tool's requirements were briefly outlined, followed by a proposal for its architecture.

From an overall perspective, we have recognised and proven the strong potential that the use of LLMs in the field of cybersecurity presents, with emphasis on automating the development and instantiation of cybersecurity artefacts, which is expected to provide a faster and more efficient response to emerging threats. Another valuable conclusion was that automating the generation of CTF exercises aims to increase the quality and quantity of practical materials on the subject of cybersecurity training. The last conclusion was that the adoption of LLM-based solutions optimises the allocation of human resources, which will contribute to a significant improvement in operational efficiency and incident response capacity.

In conclusion, we have endeavoured to provide a valuable insight into the potential future applications of LLMs in cybersecurity, emphasising the need for continuous research and improvement in the fields studied.

Appendix A: Guidelines to replicate work

This guideline aim to log the used methods to execute the prompts in both evaluation frameworks, allowing the reader to replicate the developed work. All prompts were executed directly using front-end chat interfaces, allocated in the internet. To interact with the GPT models were used the ChatGPT[4], while the Gemini model was used the Google's official chat, also entitled Gemini[5]. The remaining models were used the Perplexity[6] interface in order to interact with the models. The datasets used in both evaluation frameworks should be attached to this paper.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10207-025-01112-1.

Author Contributions Tiago Conceição wrote the main manuscript text. Nuno Cruz provided supervision to all the work performed. All authors reviewed the manuscript.

Funding Open access funding provided by FCT|FCCN (b-on). This research was supported by Fundação para a Ciência e a Tecnologia, through LASIGE Research Unit, ref. UID/00408/2025.

Data Availability Data is provided within the supplementary information files.

Declarations

Competing interest The authors declare that they have no competing interest as defined by Springer, or other interests that might be perceived to influence the results and/or discussion reported in this paper.

Competing interests The authors declare no competing interests.

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[4] https://chatgpt.com/

[5] https://gemini.google.com/

[6] https://www.perplexity.ai/

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